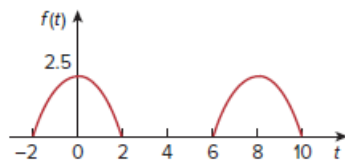


Problem

Determine the Fourier coefficients a_n and b_n of the first three harmonic terms of the rectified cosine wave in Fig. 17.52.

Figure 17.52:



Step-by-step solution

Step 1 of 6

Refer to waveform shown in Figure 17.52 in the textbook.

The function $f(t)$ is,

$$f(t) = \begin{cases} 2.5 \cos\left(\frac{\pi t}{4}\right) & -2 < t < 2 \\ 0 & 2 < t < 6 \end{cases}$$

The fundamental frequency is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{8} \\ &= \frac{\pi}{4} \end{aligned}$$

Since, the function is even symmetry; the value of coefficient b_n is,

$$b_n = 0$$

Therefore, the first three harmonics of coefficient b_n are,

$$\begin{bmatrix} b_1 = 0 \\ b_2 = 0 \\ b_3 = 0 \end{bmatrix}$$

Step 2 of 6

Calculate the dc component of the a signal.

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T f(t) dt \\
 &= \frac{1}{8} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) dt + \int_2^6 (0) dt \right] \\
 &= \frac{1}{8} \left[\frac{2.5 \sin\left(\frac{\pi t}{4}\right)}{\frac{\pi}{4}} \right]_{-2}^2 + 0 \\
 &= \frac{10}{8\pi} \left[\sin\left(\frac{\pi(2)}{4}\right) - \sin\left(\frac{\pi(-2)}{4}\right) \right] \\
 &= \frac{10}{8\pi} \left[\sin\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right) \right] \\
 &= \frac{10}{8\pi} [1+1] \\
 &= \frac{10}{4\pi} \\
 &= 0.7958
 \end{aligned}$$

Therefore, the value of dc component, a_0 is 0.7958.

Step 3 of 6

Calculate the value of coefficient a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t dt \\
 &= \frac{2}{8} \int_0^8 f(t) \cos n\omega_0 t dt \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos n\left(\frac{\pi}{4}\right) t dt + \int_2^6 (0) \cos n\omega_0 t dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{n\pi t}{4}\right) dt \right] \dots\dots (2)
 \end{aligned}$$

Step 4 of 6

Calculate the first harmonic of coefficient a_n .

$$\begin{aligned}
 a_1 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{(1)\pi t}{4}\right) dt \right] \\
 &= \frac{2.5}{4} \left[\int_{-2}^2 \cos^2\left(\frac{\pi t}{4}\right) dt \right] \\
 &= \frac{2.5}{4} \left[\int_{-2}^2 \frac{1 + \cos\left(\frac{\pi t}{2}\right)}{2} dt \right] \\
 &= \frac{2.5}{8} \left[t + \frac{\sin\left(\frac{\pi t}{2}\right)}{\frac{\pi}{2}} \right]_{-2}^2 \\
 &= \frac{2.5}{8} \left[2 - (-2) + \frac{\sin(\pi) - \sin(-\pi)}{\frac{\pi}{2}} \right] \\
 &= \frac{2.5}{8} \left[2 + 2 + \left(\frac{2}{\pi}\right)(0 - 0) \right] \\
 &= 1.25
 \end{aligned}$$

Therefore, the value of first harmonic of coefficient a_n is 1.25.

Step 5 of 6

Calculate the second harmonic of coefficient a_n .

$$\begin{aligned}
 a_2 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{(2)\pi t}{4}\right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{\pi t}{2}\right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \left(\frac{\cos\left(\frac{\pi t}{4} + \frac{\pi t}{2}\right) + \cos\left(\frac{\pi t}{4} - \frac{\pi t}{2}\right)}{2} \right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 1.25 \left[\cos\left(\frac{3\pi t}{4}\right) + \cos\left(-\frac{\pi t}{4}\right) \right] dt \right] \\
 &= \frac{1.25}{4} \left[\frac{\sin\left(\frac{3\pi t}{4}\right)}{\frac{3\pi}{4}} + \frac{\sin\left(\frac{\pi t}{4}\right)}{\frac{\pi}{4}} \right]_{-2}^2 \\
 &= \frac{1.25}{4} \left[\frac{\sin\left(\frac{3\pi(2)}{4}\right)}{\frac{3\pi}{4}} + \frac{\sin\left(\frac{\pi(2)}{4}\right)}{\frac{\pi}{4}} - \frac{\sin\left(\frac{3\pi(-2)}{4}\right)}{\frac{3\pi}{4}} - \frac{\sin\left(\frac{\pi(-2)}{4}\right)}{\frac{\pi}{4}} \right] \\
 &= \frac{1.25}{4\pi} \left[-\frac{4}{3} + 4 - \frac{4}{3} + 4 \right] \\
 &= 0.5305
 \end{aligned}$$

Therefore, the value of second harmonic of coefficient a_n is 0.5305.

Step 6 of 6

Calculate the third harmonic of coefficient a_n .

$$\begin{aligned}
 a_3 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{(3)\pi t}{4}\right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \cos\left(\frac{\pi t}{4}\right) \cos\left(\frac{3\pi t}{4}\right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 2.5 \left(\frac{\cos\left(\frac{\pi t}{4} + \frac{3\pi t}{4}\right) + \cos\left(\frac{\pi t}{4} - \frac{3\pi t}{4}\right)}{2} \right) dt \right] \\
 &= \frac{1}{4} \left[\int_{-2}^2 1.25 \left[\cos(\pi t) + \cos\left(-\frac{\pi t}{2}\right) \right] dt \right] \\
 &= \frac{1.25}{4} \left[\frac{\sin(\pi t)}{\pi} + \frac{\sin\left(\frac{\pi t}{2}\right)}{\frac{\pi}{2}} \right]_{-2}^2 \\
 &= \frac{1.25}{4} \left[\frac{\sin(2\pi)}{\pi} + \frac{\sin(\pi)}{\frac{\pi}{2}} - \frac{\sin(-2\pi)}{\pi} - \frac{\sin(\pi)}{\frac{\pi}{2}} \right] \\
 &= \frac{1.25}{4\pi} [0 + 0 - 0 - 0] \\
 &= 0
 \end{aligned}$$

Therefore, the value of third harmonic of coefficient a_n is $\boxed{0}$.